AOU
الجامعة العربية المفتوحة
Arab Open University
Faculty of Computer Studies

TM271 Machine Learning and Deep Learning

Midterm Examination (MTA)
Spring Semester 2024/2025

Date: 09 /April/2025

Number of Exam Pages: 2 (including this cover sheet)	**Time Allowed: 120 Minutes**

Instructions:

- Total Marks: 30
- This exam consists of **3** parts.
- **ALL questions** must be answered in the External Answer booklet.
- Be sure you write your **name and ID** on the External Answer booklet.
- **Calculators** are **not** allowed.

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Part – A (5 Marks)

Question 1) Select the correct choice from the given multiple choices.

a) Which metric is generally preferred for evaluating classification models when dealing with balanced datasets?

A) Accuracy
B) Precision
C) Recall
D) All of the above

b) In classification tasks, what do traditional decision trees predict?

A) Discrete class labels
B) Discrete class probabilities
C) Continues Values
D) Cluster memberships

c) What is the purpose of using a test set during model training?

A) To train the model further
B) To evaluate model performance over unseen data
C) To add more training examples
D) To decrease model complexity

d) Which technique is commonly used to reduce the number of features in a dataset while preserving important patterns?

A) PCA
B) K-Means
C) Random Forest
D) Logistic Regression

e) In clustering, what is the main task of the model?

A) Distance between data points
B) Minimize intra-cluster and maximize inter-cluster distance.
C) Variance within clusters
D) Euclidean distance between centroids

Part – B (15 Marks) Provide brief answers to the following questions.

Question 2) What are the supervised machine learning algorithms? Explain any one in detail (3 Marks)

Question 3) What is confusion matrix? How does it evaluate performance of classifier? What metrics can be evaluated using confusion Metrix. (3Marks)

Question 4) What is the role of cross-validation during hyperparameter tuning? (3Marks)

Question 5) What are the Non-Linear Dimensionality Reduction Methods? Explain any two methods. (6 Marks)

Part – C (10 Marks) This part does not need any programming.

You are working as a data analyst for a financial services company that wants to segment customers based on their financial behavior. The company aims to identify different customer groups based on their credit card usage, spending habits, repayment patterns, and risk levels. The goal is to develop personalized financial products and risk assessment models tailored to each segment.

Instructions:

Question 6) Define the problem and choose appropriate machine learning techniques. (4 marks)

Question 7) Write the stepwise solution for the technique that you suggest in question no. 6. (6 Marks)

End of questions

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